Paper:

Common Sense from the Web? Naturalness of Everyday Knowledge Retrieved from WWW

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This paper is to suggest opportunities for advanced systems hiding in the millions of WWW pages. While usually the Internet is used for achieving knowledge for humans, we present opposite approach where a machine retrieves usual knowledge about humans, their common behaviors and feelings. We claim that in long run such capability will be necessary for every machine interacting with a human user. We will concentrate on our theories and illustrate them with the results of web-mining experiment.

Keywords: human-machine interaction, common sense, web-mining

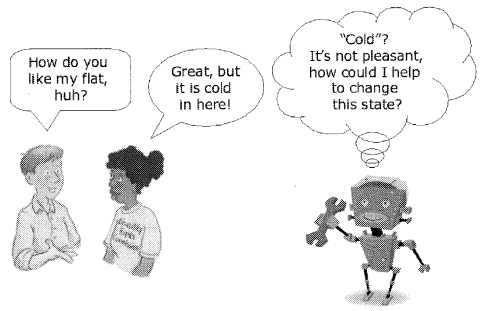


Fig. 1. An example of a situation where robot tries to deduce his possibilities while listening to users' talk.

1. Introduction

Computing with language is what we do already. However these are us who must adapt. We created simplified artificial languages to give orders to machines and to be understood without problems. Half of century later we realize like never before that not only white-dressed specialists in big rooms with computers of a size of an elephant need to give orders to machines. In twenty first century teenagers communicate with their cell phones and elderly people with microwave ovens to use them everyday. While an oven have only several options to be programmed, nowadays cell phones become personal computers which we carry almost always with us. Such machine literally becomes our partner and we would like to have an opportunity to pass our more and more complicated needs to it. Saying "from now on show this picture every time John calls me, OK?" would be much easier than traversing through menus of settings especially if you just changed your model to a new one. And we are not that far from such systems as ovens, cellular phones or car navigation systems still have limited, though expanding, purposes. Although when intelligent houses or housework robots will be developed, a number of commands for every possible situation will be enormous and knowing all the functions of our mechanical partners might be close to impossible. We will need systems which can suggest things and take some initiative, and which will have an idea about what our actions mean to us, how worthy are things to us. What are the consequences of our actions and machine's actions when it decides to take some initiative. This is knowledge that we usually call "common sense" and artificial scientists have always repeated that this might be indispensable part of many systems coping with everyday life (see **Fig.1** for an ideal housework robot guessing our needs and **Fig.3** for an algorithm example where robot self-checks its possibilities).

1.1. Common Sense Processing Today

It is obvious for everyone that machines would be regarded much more clever if they could process information while having the common sense background – the knowledge which humans are gaining especially through the first decade of their lives. The amount of data that should be inputted for achieving an universal abilities to reason in the real world is so enormous that the common sense researchers always had to limit their systems. Even if the basics for commonsensical behavior processing were made 30 years ago by – among others – Minskian frames [1], Schankian scripts [2] or casual theories of Fillmore [4], the realization of universal program that would behave naturally in any environment with any user is still technically impossible. And in authors' opinion, not universal commonsensical knowledge cannot be named "commonsensical" because of it limitations in usage. When the next decades gave us thousands times

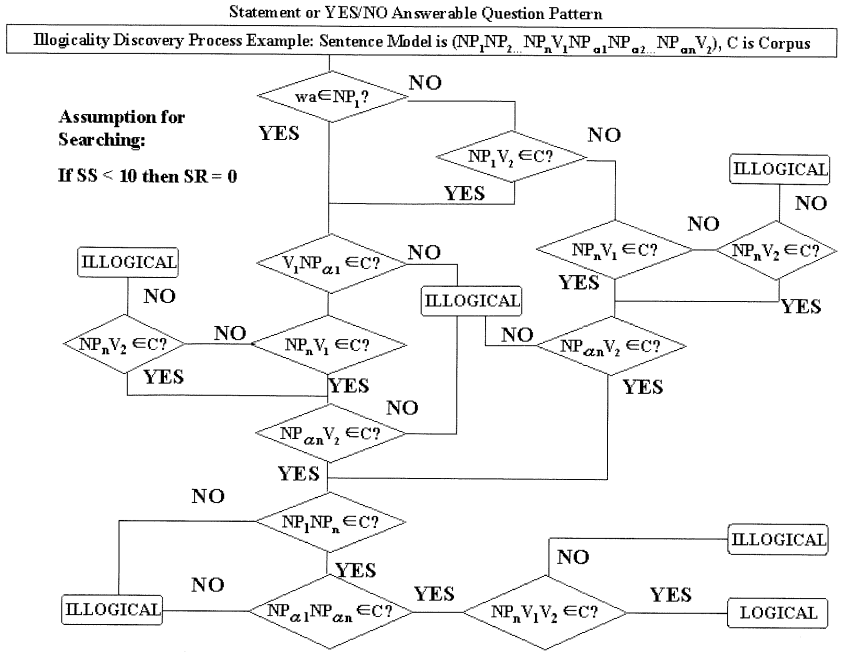


Fig. 2. The algorithm discovering Japanese sentence illogicality in the Internet corpus; SS=Searching String, SR=Search Result.

faster computers, most of Artificial Intelligence proponents were sure that human-level intelligence is not so far from the achievement. Then the same researchers were alarming that it is much more difficult task than they were presuming [5–7] and it became obvious that the calculation power may be enough but the computer scientists understood that they still lack of data. About the same time, the biggest source of data today – the Internet – started gaining its popularity among the universities and scientific laboratories around the world. The nineties of the previous century brought much broader band and bigger hard disks what had its reflection on the corpora processing and the spread of World Wide Web which also became a source for Natural Language Processing (see **Fig.2** for an example how the text statistics and search engine can solve NLP problems instead of other methods and corpora). However, from their beginning, the Internet resources were treated as a huge informational garbage dumping ground where one must use sophisticated methods to retrieve data that are useful in the trade purposes meaning. The beginning of this century showed us quite a big range of applications that are using the WWW as a corpus [8, 9] but there is still one fundamental difference – humans retrieve information to make them richer or smarter but they might retrieve other kind of knowledge which could be helpful to make the machines smarter. We suggest that knowledge which is abandoned as useless by others could become a crucial part of the "near future" applications mentioned by Rzepka et al. [10, 11]. In this paper we confirm the naturalness of data retrieved while automatically creating Schankian scripts. As far as we know, our work is the only attempt to achieve a automated Schankian scripts creation.

1.2. Intelligence from the Web

"Intelligent Web" gathers more and more interest around the world and consists of many fields as Web Mining, Weblog Mining, Web Agents, Web Farming, Intelligent Information Extraction, and many others where WWW becomes a source for retrieving information, representing knowledge or gathering parallel examples for machine translation [12]. Main stream is to make the vast abyss of WWW more computer-readable through formalizing ontologies and creating semantic webs build upon strict rules. However, our approach is based on the weakest type of ontologies – the raw text. The reason why we use text from the Web is an easy access to statistic information on texts which are "alive" – being updated every day and still have informal style. For example blogs contain more everyday knowledge than newspaper corpora what we already proved [22]. There are two famous examples of building ontologies for common sense which are CyC [13] and OpenMind CommonSense [14]. The Cyc project has its own ontology language called CycL, based on first-order predicate calculus with some higher-order extensions and OMCS uses Internet users to gather expressions describing common sense. In our opinion the former needs too much money and is to laborious when the latter lacks automatic approaches as we claim that it is very difficult to input commonsense knowledge manually. OMCS also largely misses a very important factor – the information existing in frequencies of expressions.

1.3. An Idea of an Average Personality

As "understanding the world" leads to "understanding a natural language", we need as much world information

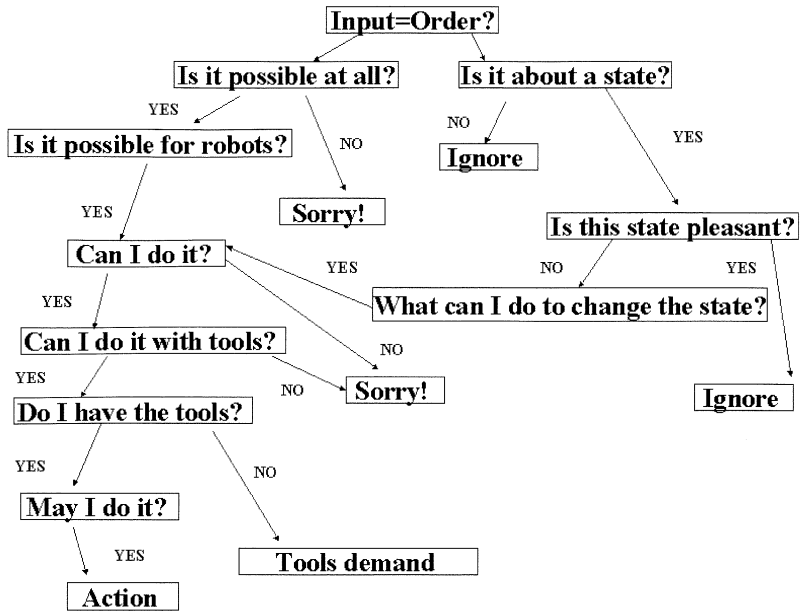


Fig. 3. The algorithm for a robot using Internet resources for proposing its own actions.

as possible. The Schankian claim is that structured knowledge dominates understanding. In authors' opinion, such structures are built by occurrences – the more times something happens, the more likely it is that we will treat it as a natural phenomenon. Some of us have very different experiences than the others but there are events that are common for everyone – even if not experienced personally – known from the experiences of others. Therefore we presume that it is possible to gain the commonsensical knowledge learning only form the experiences of others. We also presume that a machine can learn the experience without experiencing them as in the case where the blind child learns about the visual world [16, 17]. The more frequent a given experience is, the more common an event can become. Hence, we assume that every human has a "general personality" which is an axis of all our behaviors. The more uncommon things we do, the farther the common sense degree deflects from the axis which could work as, for example, a safety device, for robots [18]. Our task is to create methods for calculating such a general personality which could be a database of facts and behaviors which are common to one culture circle like a Japanese one used in our research. We already provided the system with general opinions [19], we decided to provide "Mr. Internet" with knowledge of basic actions patterns influenced by works of Schank.

1.4. About Japanese Language

The basic Japanese word order is Subject Object Verb. Subject and object are usually marked by particles which come after the noun and they indicate relation between nouns and usually verbs often taking the role of English prepositions (See **Table 1**). The basic sentence structure is topic-comment. For example, "Kochira-wa Suzuki-san desu". "Kochira" ("this") is the topic of the sentence, indicated by the particle wa. The verb function is taken by "desu" (which might be translated to "is"). As a phrase, "Suzuki-san desu" is the comment. The loose translation could be "As for this person, (it) is Mr./Mrs./Ms. Tanaka". Thus Japanese is often called a topic-prominent language, which means it indicates the topic separately from the subject, and the two do not always coincide which is useful during web-mining but becomes a hard task when one creates a talking system. The sentence "Daidokoro-de sara-o arau" means "I/you/he/she/it/we/they wash dishes in the kitchen". We chose this structure for experiments as a basic and most common sentence structure that must be dealt with by a housework robot (performing some task at some place). Japanese language can be written in four ways – using Chinese ideograms (Kanji), two kinds of syllables sets (Kana) and latin characters.

Table 1. Examples of Japanese Particles.

Particle	Role
WO	Object-Indicating
NI	Direction-Indicating
DE	Place or Means of Action-Indicating
HE	Destination-Indicating
TO	Connective
GA	Linking-Indicating
WA	Topic-Indicating
MO	Addition-Indicating
DEMO	Emphasis-Indicating

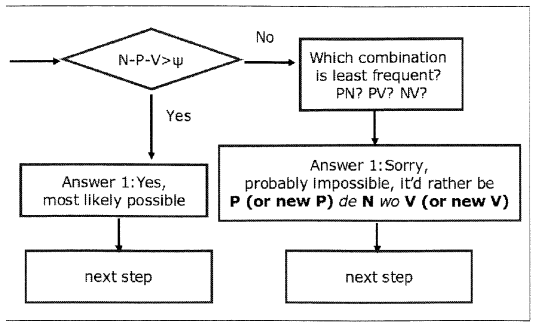


Fig. 4. The first step: confirming the possibility.

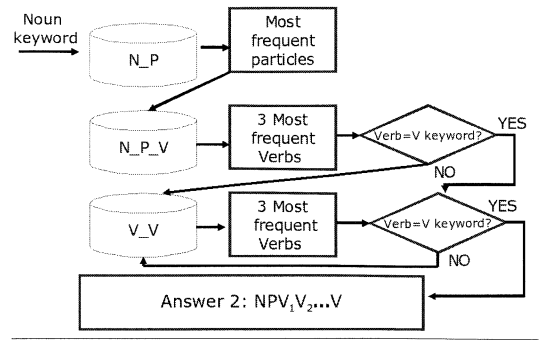


Fig. 5. The second step: checking how usual the action is.

2. Algorithms

We have been performing trials, frequency checks and experiments on various grounds of common sense knowledge. Here we introduce the latest fields which we work on – recognition of possibility, minimal Schankian scripts and proposals in case of errors (verbs only so far). By minimal Schankian script we mean the simplest *action*, which is usually defined as "something a person can do". Therefore "throwing a ball" we treat as a minimal description of an action which can be further broaden, to e.g. "strong throwing of a ball at the wall by a boy" which is basically the same action as "strong throwing of a ball at the window by a girl" but can have diametrically different consequences.

2.1. Input Sentence Analysis

This algorithm analyses one pattern sentences describing simple actions in form of one noun, one particle and one verb (as mentioned in the section about Japanese language) which were freely entered through WWW interface. Below is the step-by-step procedure (See **Figs.4** and **5**):

1. (Place)-Noun DE Noun WO Verb [input sentence]
2. Is it possible?
   if YES then NEXT
   if NO then SEARCH A CASE WHEN YES (e.g.: if place A is impossible suggest up to 3 most likely places B)
3. ChaSen [20] extracts nouns, verbs, particles and adjectives
   a) creating Noun Dictionary
   b) creating Noun-Particle Dictionary (o, wa, ga, no, ni, de, he, mo, to, kara. See: **Table 1**.)
   c) creating Verb Dictionary 1 (in dictionary form *jisho-kei*)
   d) creating Verb Dictionary 2 (in form which was actually used)
   e) creating Verb-Verb Dictionary
   f) creating Noun-Particle-Verb Dictionary
   g) creating Adjective Dictionary
   h) creating Places Dictionary

2.2. Schankian Scripts

2.2.1. The Theory

In his classic book [2], Schank has introduced inquiry into the nature of knowledge that is needed to understand the world and understand natural language. His main claim is that structured knowledge dominates understanding while the question is the content of these structures. He takes a pragmatic approach that does not separate form from content. He proposes conceptual primitives we decided to partially adopt although his proposal has not been tested psychologically. Schank concentrates on memory, and in particular memory organization but we use only our original Bacterium Lingualis Memory concept [3], which will not be mentioned here. He argues, and we agree with him, that understanding language involves causally connecting thoughts/sentences. Because causality is often implied or incompletely described, it is usually harder to understand connected text than individual sentences. Therefore, he describes a formal representation (causal syntax) of causal chains which is to the discourse level what Conceptual Dependency is to the sentence/thought level. Rules are provided in which every primitive action is associated with the set of states it can affect as well as those that enable it. Schank describes scripts as groups of causal chains that represent knowledge about frequently experienced events (the most famous example is "going to a restaurant"). In other words, a script is a stereotyped sequence of actions that defines a well-known situation and has associated with it: a number of roles for the actors (different points of view on the situation, e.g. customer vs waiter vs cook), different tracks (e.g. restaurant, fast-food), different scenes (e.g. enter, order, eat, pay); each scene has a main conceptualization, which must have happened if the scene is instantiated, as well as props, entry conditions, results, branches and loops etc.

2.2.2. The Practice

Scripts are important part of our basic thinking processes and can be triggered as associations to any word we are told. This supposed that this is the first step for creating machine imagination. We also perform experiments on causal dependencies (IF-forms), which are retrieved in

almost identical way as scripts. Such a system's architecture can be summarized into the following processing steps:
- A noun is assigned for a **keyword** (any keyword spotting method can be applied);
- The system uses our web corpus for frequency check to retrieve **3 most frequent verbs** following the keyword noun;
- The **most frequent particle** between noun keyword and 3 most frequent verbs is discovered;
- For creating bi-gram the system retrieves a list of most frequent **verbs** occurring **after** the previously chosen **verb**;
- By using Google search engine [21], the system checks if the noun-particle unit occurs with new verb-verb unit for **time-sequence actions** and verb-if unit for **casual dependencies**;
- If yes – the VO-then-V and VO-if-VO **units are stored**:

$\mathbf{VO_{then}V} = \mathbf{N} + \mathbf{P}_{max} + \mathbf{V}_{max1} + \mathbf{V}_{max2}$
$\mathbf{N}$: *Triggering noun (keyword);*
$\mathbf{P}_{max}$: *most frequent particle joining noun and verb;*
$\mathbf{V}_{max1}$: *most frequent verb occurring after the* $\mathbf{N}$*;*
$\mathbf{V}_{max2}$: *most frequent verb occurring after* $\mathbf{V}_{max1}$*;*

$\mathbf{VO_{if}V} = \mathbf{N_1} + \mathbf{P}_{1max} + \mathbf{V}_{1max} + if + \mathbf{N_2} + \mathbf{P}_{2max} + \mathbf{V}_{2max}$
$\mathbf{N_1}$: *Triggering noun (keyword);*
$\mathbf{P}_{1max}$: *most freq. particle joining first noun with a verb;*
$\mathbf{V}_{1max}$: *most freq. verb after the* $\mathbf{N_1P_1}$*;*
$\mathbf{N}_{2max}$: *most freq. noun after* $\mathbf{N_1P}_{1max}\mathbf{V}_{1max}$ *and "if";*
$\mathbf{P}_{2max}$: *most freq. particle joining* $\mathbf{N_2}$ *and* $\mathbf{V_2}$*;*
$\mathbf{V}_{2max}$: *most freq. verb after* $\mathbf{N_2P_2}$*;*

To retrieve minimal scripts (mentioned in the Theory Section) consisting of at least two actions following each other, we created a program which performs following steps:

- Search the most frequent particle following noun keyword in Noun-Particle Dictionary;
- Search three most frequent verbs from Noun-Particle-Verb dictionary as the first actions using Noun keyword and the most frequent particle;
- if the Verb of Noun-Particle-Verb is Verb keyword, the script is a minimal script; if not, search the most three nouns using the verbs from Noun-Particle-Verb dictionary and search the most three verbs from Verb-Verb Dictionary as the second action using the first verbs.
  if the verb is not followed by any actions, the script is be deleted;
- repeat previous step until the last action is verb keyword.

In this paper we will describe experiments for the latter algorithm for minimal scripts as the results for the two mentioned above were already presented [22, 23].

Table 2. Naturalness of Schankian scripts and strings for their creation made out of 36 sentences inputted by users.

Number of Evaluated Strings	Natural	Unnatural
600	571	29
100%	95.15%	4.83%

3. Experiments and Results

We created cgi-based WWW interface where users (which were authors during the experiment: Polish, Chinese, Japanese, one female and two males of average age 35.6) are able to freely input place nouns, nouns and verbs while we limited particles to place indicating and object indicating ones (for the reasons mentioned in the Section explaining Japanese language). Place indicating particle is also tool indicating particle therefore we did not suggest that first noun must be a place noun. Basically we wanted to confirm naturalness of frequencies retrieved by using our own web-based dictionaries for shorter combinations and Google Search Engine for longer sets of words. However, we also checked the ability of the algorithm to retrieve actions (verbs only, though) following one which is inputted to create a script made of three, not two as before, actions. As a rule, we counted repeated actions as natural if only they were not repeated one after another, for example "watch – think – watch (again)" or "drink – go back home – drink (again)" were accepted while "look – look – look" or "come back home – watch – watch" were not. Such repetitions were due to the fact that our system still does not retrieve noun-particle units with verbs retrieved for actions following input action. This was also the reason we are still not able to fairly evaluate scripts retrieval. If the noun was common topic for actions or it was not needed, they were natural, for example "Give birth to a child – take care of (it) – do best", "wait for friends – have fun – go back home", "wait for friends – go back home – sleep" or "take out the trash – look through (it) – find out (something)". We also approved grammatical forms meaning movement or giving/receiving (*-te morau/ageru, -te kuru/kaeru/iku*) but we did not approved continuous *-te iru* and *-te miru* (try to...) as static and not associated with actions in a satisfactory level. When evaluating naturalness of retrievals we were labeling natural sentences with high frequency and unnatural sentences with low frequency as natural and opposite. Border ψ[1] was set 10 for scripts with two and more actions and 100 for strings with one action only. After evaluating 600 retrievals we found 29 unnatural frequencies and other errors like the same verbs written in different characters (Kanji/Kana), which resulted in surprisingly low error rate 4.83% (See **Table 2**). However, it must be mentioned here that for example metaphors like "eating trash" were automatically evaluated as natural by a human but might become a bigger problem to a literal machine.

1. This border is usually calculated from the dependency between searched query length and the frequency in a search engine – the longer the query is the lower ψ becomes.

4. Discussion

Interestingly, the errors in Japanese grammar, appearing due to the fact that also non-native Japanese speakers were taking part in our experiment, were correctly discovered and even corrected in the cases when the noun-verb pair was frequent, e.g. "growing flowers" inputted as *hana-wo sodatsu* were answered with a correct suggestion *hana-wo sodateru*. This time we concentrated on verbs, the next step will be noun suggesting ability. The last step will be learning the emotional load of adjectives in a given situation. We think that machines should not only recognize errors of humans but also fill the blanks with usual items, places, actions or descriptions. If it is possible, receiving only "go and buy it" would be enough for a machine to add missing information about items needed to be used or places which should be set for destination without asking its user about every single detail. The next two barriers awaiting us is a "robot personality" and "fiction discovery". Our idea of Average Personality is useful for a robot to understand human but it does not say much about machines possibilities. Can machine do more than its engineer set it up for? For example it can "pick up", "push" or "drop" things but some combinations of this simple functions might lead to more sophisticated ones. How a machine could guess what other machines can do? Why not by asking a search engine if other robots / machines / computer can do it? One may say "impossible because there is too much SF out there". We agree with it and have already started working on an algorithm which could learn how to tell a fictional story from a true one.

5. Future Work

We plan to experiment on many aspects of a very wide topic called "Commonsense Knowledge", although in this step we try to cover only simple chains of actions. We agree that everyday knowledge depends on many factors working or not working in given situations. We have already proved that common sense retrieved from WWW depends on the culture [24] – this is why we work on languages closely associated with one culture as Japanese, Polish or Chinese (the only restriction while searching the Web is language). We presume that scripts retrieved in such a language should be useful in the environment of its users. On the other hand, an original Schankian script for a restaurant cannot be used in fast food bar in any culture but WWW-based automatic approaches are able to discover that, for instance, the actor WAITER is missing and the script should be modified. We also agree that understanding one piece of common sense might lead to understanding another, which is not written. We are very interested in such mechanisms and inductive learning of "tacit knowledge" based on already achieved data will be one of our next steps.
We are quite convinced that the only fair evaluation of common sense processing is to evaluate not the knowledge itself but how it was used. For that reason we plan to build a housework robot simulation system which will give us an opportunity to try the usefulness of the surface language structures and their statistics in a given situation and place, with specified tools or actors. We assume that for deciding or suggesting its own actions (building missing ontologies automatically), the robot with default (background and domain) knowledge should also have an ability to decide if the state of user or the place is negative or positive. We think that guessing such emotional information from users expression can be done without deep semantic analysis as well as retrieving knowledge about actions and their consequences. Such knowledge could be used later as a support for more sophisticated systems as decision makers or Human-Computer Interfaces, but our future goal is to find an universal method of retrieving very obvious (for humans) truths out of unannotated text corpora.

6. Conclusion

In this paper we introduced our theories and ideas about how the Internet text resources may be helpful for automatizing of a very laborious task of inputting commonsensical knowledge needed by machines for better human-machine interactions. By retrieving associations, correlations, causal relations and reusing classic thoughts of cognitive science and artificial intelligence like scripts or frames we shall be able to provide machines with the knowledge we are achieving from the day we are born without even realizing it. We also showed that web pages indexed by nowadays search engines are enough to give us word combinations frequency results evaluated natural in 95.15% of 600 counts performed on randomly inputted data. We claim that without the common sense knowledge a machine will not be able to program itself according to our sophisticated demands passed in a natural language. Therefore we think the Internet gives us many opportunities in achieving this goal by, among others, the means suggested in this work.

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Appendix A. Evaluation Examples

Examples showing evaluation process (majority decision) and retrieval outputs in original:

Ribingu-de gohan-wo taberu (to eat meal at the living room): Previous: jiru-wo kakete (after putting broth on) [OK] Next: okawari-ga moru (to get a refill) [OK]
Ribingu-de kome-wo taku (to boil rice at the living room) Previous: mizu-wo irete (after pouring water in) [NG] Next: gohan-wo taberu (to eat a meal) [OK]
Hoteru-de tabako-wo suu (to smoke a cigarette at a hotel) Previous: hi-wo tsukete (after putting up light) [OK] Next: niou (to smell) [OK]
Doko-ka-de gomi-wo taberu (to eat trash somewhere) Previous: kuchi-ni irete (after putting into his/her mouth) [OK] Next: sakana-wo shinu (get fish dead) [NG]
Mise-de ranchi-wo taberu (to eat out) Previous: soosu-wo kakete (after putting sauce on) [OK] Next: ie-ni kaeru (to go back home) [OK]
Toire-de mado-wo shimeru (to close a window at the toilet) Previous: kaze-wo hiite (after getting a cold) [OK] Next: heya-de neru (to sleep at a room) [OK]
Doko-ka-de gamu-wo kamu (to chew a gum somewhere): Previous: wakkusu-wo kakete (after putting some wax on) [NG] Next: shokuhin-wo taberu (to eat a product) [OK]
Michi-de gomi-wo hirou (to pick up trash at the street): Previous: hitotachi-ni nite (after being similar to people) [NG] Next: ie-ni mochikaeru (to brig it back home) [OK]
Izakaya-de o-sake-wo kobosu (to spill out alcohol at the bar): Previous: ojisan-ni kagitte (only some fellows) [NG] Next: ie-ni kaeru (to go back home) [OK]
Hikouki-de kodomo-wo umu (to give birth at the aeroplane): Previous: aite wo erande (after choosing a partner) [OK] Next: kodomo-wo sodateru (to bring up a child) [OK]

Appendix B. Input Sentences

Sentences (alphabetical order) inputted for analysis (tools instead places and wrong grammar were allowed as nouns and verbs were free to choose)

Chuusha-jou-de tomodachi-wo matsu: Waiting friends at a parking lot
Daidokoro-de kudamono-wo kau: Buying fruits at the kitchen
Daidokoro-de reizouko-wo arau: Washing a refrigerator at the kitchen
Daidokoro-de sentaku-wo hosu: Drying laundry at the kitchen
Daidokoro-de tabako-wo suu: Smoking cigarettes at the kitchen
Doko-ka-de gamu-wo kamu: Chewing a gum somewhere
Doko-ka-de gomi-wo taberu: Eating trash somewhere
Eigakan-de eiga-wo miru: Watching a film at the movies
Eigakan-de osake-wo kobosu: Spilling alcohol at the movies
Eigakan-de osake-wo nomu: Drinking alcohol at the movies
Heya-de gohan-wo taberu: Eating a meal at the room
Heya-de mado-wo hirakeru: Opening window at the room
Heya-de mado-wo shimeru: Closing window at the room
Heya-de o-mizu-wo taberu: Eating water at the room
Hikouki-de kodomo-wo umu: Giving birth inside an aeroplane
Hoteru-de tabako-wo suu: Smoking cigarettes in a hotel
Izakaya-de eiga-wo miru: Watching a film at the Japanese-style bar
Izakaya-de o-sake-wo kobosu: Spilling alcohol at the Japanese-style bar

Kichin-de karee-wo kau: Buying curry at the kitchen
Konbini-de o-bentou-wo kau: Buying a lunch-box at convenient store
Michi-de gomi-wo hirou: Picking up trash on the street. Mise-de ranchi-wo taberu: Eating lunch in a bar
Mise-de tamago-wo kowasu: Breaking an egg at a store
Pasokon-de shinbun-wo yomu: Reading a newspaper with computer
Reizouko-de hon-wo arau: Washing a book in a refrigerator
Resutoran-de o-sake-wo nomu: Drinking alcohol at restaurant
Ribingu-de kome-wo taberu: Eating raw rice at living room
Ribingu-de kome-wo taku: Boiling raw rice at living room
Ribingu-de gohan-wo taberu: Eating boiled rice at living room
Sukii-jou-de shawaa-wo abiru: Taking shower at the skiing ground
Suupaa-de yasai-wo toru: Picking up vegetables at a store
Toire-de gamu-wo kamu: Chewing a gum at the toilet
Toire-de gamu-wo suteru: Throwing away a gum at the toilet
Toire-de gomi-wo suteru: Throwing away trash at the toilet
Toire-de mado-wo shimeru: Closing a window at the toilet
Uchuu-de hana-wo sodatsu: Growing flowers in space



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